



Multi-Chain NFT Management Platform

A single Web3 platform to mint, manage, and monetize NFTs across multiple blockchains, wallets, and marketplaces — one account, one unified portfolio, one dashboard.

INDUSTRY

ENGAGEMENT

DELIVERY MODEL

PLATFORM

Web3 · NFT · Fintech

~8-month MVP

Agile MVP

Serverless AWS +
Web3

Confidentiality. Sanitized portfolio case study. The client and product names, founder/partner identities, proprietary business rules, and internal source are withheld under NDA. Screenshots are from a demo environment with the product logo redacted and illustrative (non-real) data; outcome figures are representative and anonymized.

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AT A GLANCE

CHALLENGE

The NFT world is fragmented across competing blockchains, wallets, and marketplaces — no unified portfolio, high technical burden to mint/move assets, and no place to see holdings, value, and transactions in one view.

SOLUTION

A serverless, multi-tenant Web3 platform: one account for many wallets and smart contracts, minting (incl. lazy-mint), a unified cross-chain portfolio, and an event-driven engine that keeps blockchain data fresh.

RESULT

A scalable MVP delivered on AWS with a rich, responsive UX — making a multi-chain NFT experience feel like one portfolio, and launched to a top-3 product-of-the-day ranking.

#3 launch

Top-3 Product of the Day on a leading launch platform

~8 months

MVP delivered end-to-end (product → architecture → launch)

Multi-chain

One account across wallets, chains & marketplaces

Serverless

Multi-tenant AWS + Web3 stack

CHALLENGE

A fragmented, multi-chain NFT experience

Value and activity in the NFT ecosystem are scattered across competing blockchain protocols (Ethereum, Polygon, Solana, Flow, Binance Smart Chain, Theta, ...), each with its own wallets, marketplaces, and tooling:

- **No single view.** NFTs live in different wallets on different chains — no unified portfolio or valuation.
- **High technical burden.** Minting, lazy-minting, importing smart contracts, and moving assets each need chain-specific know-how.
- **Fragmented activity.** Buying, selling, sending, and tracking means juggling multiple wallets, marketplaces, and explorers.
- **No portfolio intelligence.** Nowhere to see holdings, value, spend, earnings, and transaction history together.

The vision: a **portfolio command-center for NFTs** — one account to manage multiple wallets and smart contracts, and streamline every NFT activity across chains behind one simple experience.

SOLUTION

One account, every chain — serverless & event-driven

A scalable, multi-tenant serverless application on AWS, integrated with Web3 infrastructure. The interactive app stays fast because all blockchain reconciliation runs in an asynchronous, event-driven pipeline.

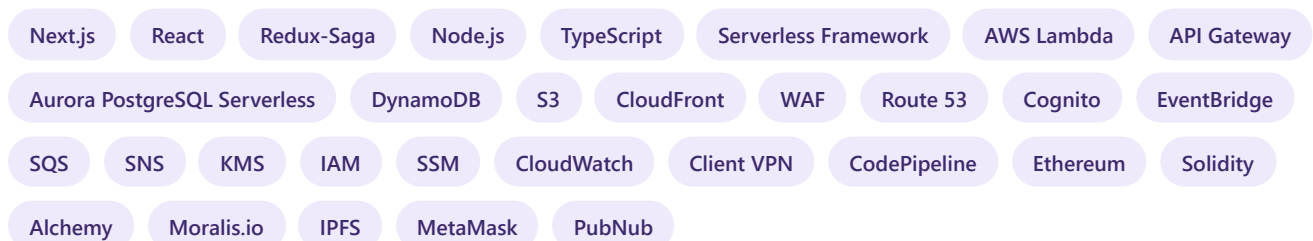
The delivery approach

- **One account, many wallets & chains:** connect multiple wallets (MetaMask) and custom smart contracts; a securely-managed platform wallet for in-app crypto operations.
- **Mint & create:** design and mint NFTs including lazy-mint (mint-on-purchase, no upfront gas) via Alchemy; assets on IPFS; custom Solidity contracts for transfer/mint/lazy-mint.
- **Unified portfolio & analytics:** all NFTs across wallets and chains in one view, with NFTs owned, portfolio value, transactions, spend/earnings, and trends — filterable by wallet, contract, and date.
- **Transact:** send/receive crypto and NFTs; streamlined buy/sell/trade.
- **Always-fresh data:** an event-driven sync engine (EventBridge → SQS → Lambda fetchers → Moralis) reconciles wallet/NFT/transaction state with the chain on a configurable schedule.

Architecture at a glance

```
Web App (Next.js) + MetaMask
| Route 53 · CloudFront · WAF · Cognito
Serverless API (API Gateway + Lambda · Node.js/TypeScript)
| Data: Aurora PostgreSQL Serverless · DynamoDB · S3
| Web3: Ethereum · Solidity · Alchemy (mint/lazy-mint) · IPFS · Moralis
| Event-driven sync: EventBridge → SQS → Lambda fetchers → Moralis → DB
Multi-tenant · private subnets (multi-AZ) · KMS · IAM · Client VPN · CI/CD
```

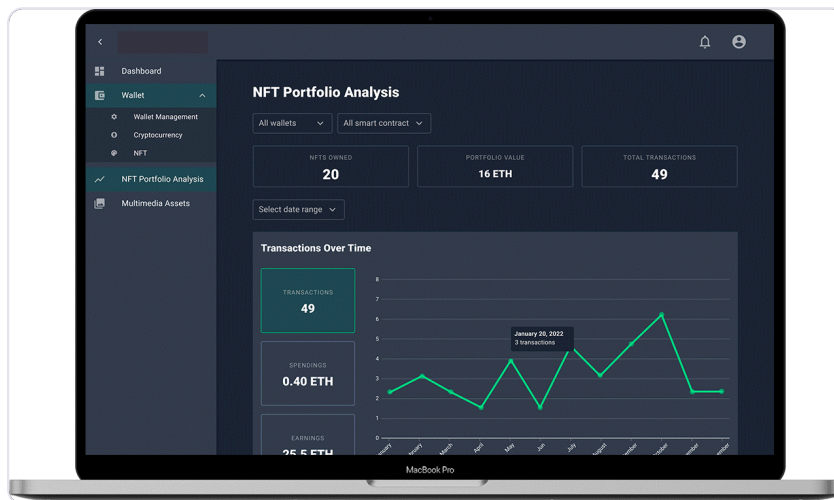
Technology



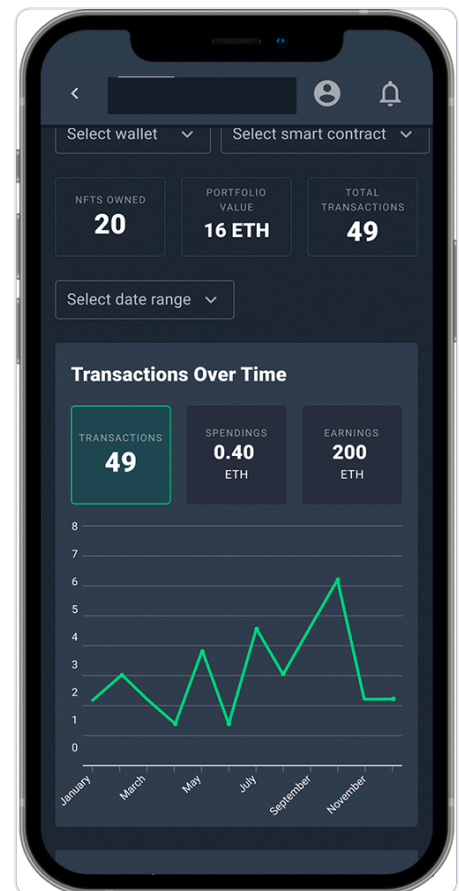
THE PRODUCT

NFT Portfolio Analysis — desktop & mobile

Demo environment. Product logo redacted; illustrative data.



Unified cross-chain portfolio — NFTs owned, portfolio value, transactions, spend/earnings, and trends, filterable by wallet and smart contract.



Fully responsive — the same portfolio on mobile.

RESULT**One portfolio — and a top-3 launch****Top-3 launch**

Ranked #3 Product of the Day on a leading launch platform

One portfolio

All NFTs, wallets & chains unified in one view

Lazy-mint

Mint-on-purchase — no upfront gas for creators

- **Fragmentation, solved.** A multi-chain NFT experience became one account, one portfolio, one dashboard — across wallets, chains, and marketplaces.
- **Rich, responsive UX.** Complex on-chain actions (mint, lazy-mint, send/receive, import contracts) made simple on desktop and mobile.
- **Scalable & secure by design.** Multi-tenant, fully serverless AWS with private-subnet compute, CloudFront + WAF, Client VPN, KMS/SSM secrets, and Cognito auth.
- **Delivered end-to-end** — product scoping, system architecture, tech-stack and Web3-partner selection, and delivery — in roughly eight months.

"[The platform] is built for NFT creators and traders to streamline all of your NFT activities — supporting all blockchains, all NFTs, in one single platform."

— Founder (identity withheld under NDA)

Note on figures. Metrics in the product screenshots are demo/sample data. Specific client results, credentials, contract addresses, and identifying details are excluded under NDA; outcomes here are described qualitatively and anonymized.